Carrol Chapter 2: Celestial Mechanics

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December 6, 2025

Definition: (Astronomical Units)

An **astronomical unit** (AU) is the average distance between the Earth and the Sun

Proposition: (Kepler's Three Laws)

1. A planet orbits the Sun in an ellipse, with the Sun at one focus of the ellipse
2. A line connecting a planet to the Sun sweeps out equal areas in equal time intervals
3. If P is the sidereal period of a planet, and a is the average distance of the planet from the Sun (in astronomical units), then they are related by
$$P^2 = a^3$$

Definition: (Ellipses)

An **ellipse** is defined by two points, called **foci**, and consists of all of the points that satisfy the following equation:

$$r_1 + r_2 = 2a$$

where r_1 and r_2 are distances from a point to the two foci, and a is the length of the **semimajor axis**, or half the length of the longest axis of the ellipse. Half of the shorter axis b is called the **semiminor axis**.

Definition: (Eccentricity)

Let d be the distance between the foci of an ellipse, and a be the semimajor axis. Then, the **eccentricity**, which measures how "stretched" an ellipse is is given by
$$e = \frac{d}{2a}$$

Definition: (Points on an Ellipse)

The **aphelion** is the point fartherst from the foci of the ellipse, where the major axis meets the ellipse.

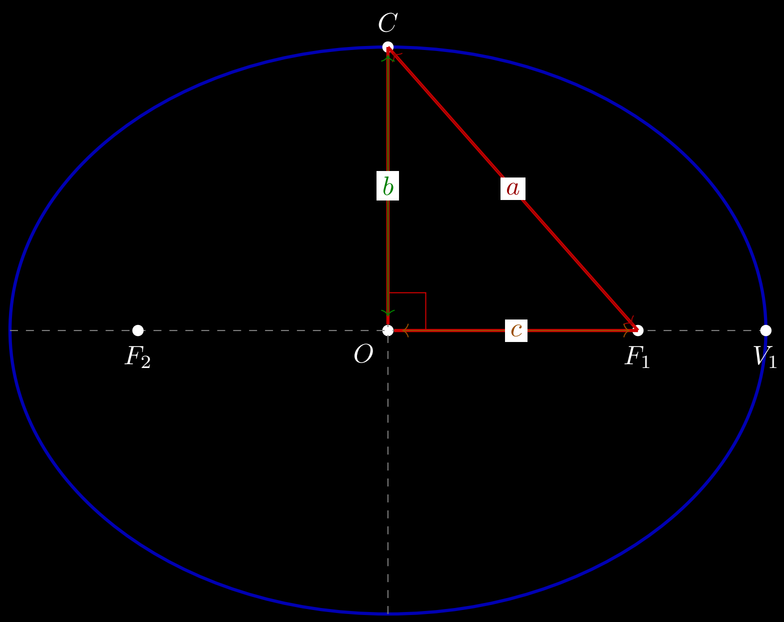
The **perihelion** is the point closest to the foci, where the minor axis meets the ellipse.

Theorem:

Let a be the semimajor axis of an ellipse, b be the semiminor axis, e be the eccentricity. Then,

$$b^2 = a^2(1 - e^2)$$

Proof:



The hypotenuse here is a since $F_1C + F_2C = 2a$, and we know that $F_1C = F_2C$. By the Pythagorean theorem, we get that

$$c^2 + b^2 = a^2$$

But by the definition of eccentricity, we know that

$$e = \frac{2c}{2a}$$

so by substituting $ae = c$, we get

$$a^2e^2 + b^2 = a^2$$

Subtracting over a^2e^2 and factoring out a^2, we get

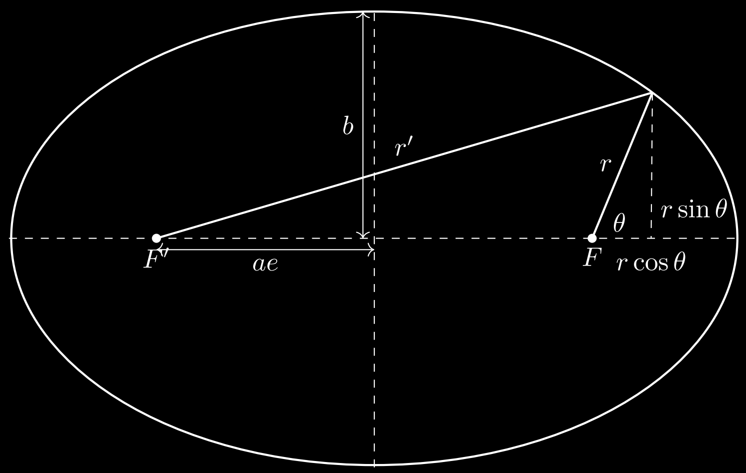
$$b^2 = a^2(1 - e^2)$$

Theorem:

Show that if a is the semimajor axis, e is the eccentricity, r is the distance from a focus to sone point p on the ellipse, and θ is the angle that line makes with the major axis, then:

$$r = \frac{a(1 - e^2)}{1 + e\cos\theta}$$

Proof:



Using the Pythagorean Theorem, we get that

$$r'^2 = r^2 \sin^2\theta + (2ae + r\cos\theta)^2$$

Then, we get

$$r'^2 = r^2 + 4ae(ae + r\cos\theta)$$

, and substituting $r + r' = 2a$ and rearranging, we get that

$$r = \frac{a(1-e^2)}{1 + e\cos\theta}$$

Exercise:

Prove that the total area of an ellipse is

$$A = \pi ab$$

Definition: (Conic Sections)

A conic section's eccentricity corresponds to the slope of the plane slicing the conic section. If $e = 1$, you get a **parabola**, given by

$$r = \frac{2p}{1 + \cos\theta}$$

where p is the singular focus of the parabola.

If $e > 1$, you get a hyperbola, given by

$$r = \frac{a(e^2-1)}{1 + e\cos\theta}$$

Proof: (Newton's Three Laws)

1. An object at rest will stay at rest and an object in motion will stay in motion unless acted upon by an external force.

$$p = mv$$

2. The net force is equal to the object's mass times acceleration.

$$F_{\text{net}} = ma$$

3. For every action there is an equal and opposite reaction.

Proposition: (Newton's Law of Universal Gravitation)

When one object's mass is much larger than the other, the orbit is roughly circular. Then, we can simplify Kepler's Third Law to read:

$$P^2 = a^3 = kr^3$$

with a constant k to account for us using meters now.

Now, note that

$$P = \frac{2\pi r}{v}$$

as the orbit's circumference divided by it's speed gives how long it takes to complete an orbit. Substituting, we get

$$\frac{4\pi^2 r^2}{v^2} = kr^3$$

Rearranging and multiplying by m yields

$$m\frac{v^2}{r} = \frac{4\pi^2 m}{kr^2}$$

But the left side is centripetal force, so

$$F_c = \frac{4\pi^2 m}{kr^2}$$

By Netwon's third law,

$$F = \frac{4\pi^2 M}{k' r^2}$$

and letting $k'' = Mk = mk'$, we get

$$F = \frac{4\pi^2 Mm}{k'' r^2}$$

and again grouping constants we get

$$F = G\frac{Mm}{r^2}$$

Theorem:

Show that gravitational potential energy is given by

$$U = -G\frac{mM}{r}$$

Proof:

By integrating force with respect to distance, we get

$$U(r) = \int G\frac{mM}{r^2} = -G\frac{mM}{r}$$